



Method of using non-specific target to complete structured light plane calibration

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ABSTRACT

The accuracy of the line structured light measurement system is determined by the calibration results of the structured light plane. Existing structured light calibration methods usually need to fabricate a specific target with high manufacturing precision to complete the calibration. In the real environment, how to calibrate the line structured light plain without a specific target becomes a key issue. In this paper, a stereo vision calibration (SVC) method is proposed and used to complete the structured light calibration with non-specific targets. It adopts the principle of stereo vision to calculate all point clouds projected on the target by structured light, and then the parameters of the structured light plane are optimized based on the probability distribution during point clouds processing. Taking 304 steel plate, aluminum round can, and book as non-specific targets, the accuracy of the plane calibrated by SVC method is measured. The minimum average measurement error is about 0.068 mm by comparing the nominal values with the measurement values of the checkerboard and different workpieces. It indicates that the SVC method can acquire the high-precision calibration with non-specific targets.

1. Introduction

Line structured light measurement is a kind of technology, which projects a line laser plane onto the surface of the object, then the coordinates of the three-dimensional point at the intersection of the light plane and the surface of the object are obtained. The line structured light measurement system has the advantages of non-contact, high speed, and high precision. Therefore, it is developing rapidly and widely used in many fields, such as railway quality inspection [1,2], industrial robots [3,4], vehicle assembly [5], and many other industrial fields. The measurement accuracy of line structured light measurement system is directly related to the structured light plane calibration [6,7]. Consequently, the accuracy of calibration should be guaranteed in line structured light measurement system.

Many scholars have conducted in-depth researches on the calibration method of structured light planes. Some methods were based on three-dimensional target. For example, Ha [8] et al. designed a three-dimensional target with multiple vertical planes, and the laser emitter position was calculated by using the points generated when the laser projected onto the calibration structure. The relative measurement errors of this method are below 0.87%. Liu [9] et al. employed spherical targets (space cone and space ball rendezvous) for structured light plane calibration, and the accuracy was about 0.09 mm. Wei [10] et al. adopted the principle of obtaining plane vanishing line with known equidistant parallel lines to complete the optical plane calibration with three-dimensional parallel line target (accuracy was about 0.12 mm). The accuracy of these calibration methods usually depends

on the production of a three-dimensional target with high manufacturing accuracy, which increases the processing cost and consume a lot of time. Thus, some researchers exploited the two-dimensional target for calibration. Wang [11] et al. used the plane checkerboard target to take photos while constantly changing the robot's attitude, and the hand-eye matrix and camera parameters were used to complete the structural light plane calibration. In this method, 16 photos under six attitudes were taken in total, and only 88 corner coordinates on the checkerboard were used to acquire the calibration accuracy (about 0.4 mm). Li [12] et al. projected structured light on the edge of the chessboard target and introduced the random sample consensus algorithm to improve the calibration accuracy to 0.25 mm. Xu [13] et al. utilized a six-axis robot to project line structured light on the marking points, and multiple calibration equations were constructed and solved based on the hand-eye calibration algorithm of the robot. They optimized the calibration equations with the least square method, and the calibration accuracy of approximately was about 0.2 mm. Wei [14] et al. installed two laser emitters on multiple degrees of freedom robotic hands and calibrated the structured light plane equation by the coordinate value of the linear structured light fringe in the camera coordinate system. The RMS error of the standard ruler measured by this method was 0.26 mm. The two methods [13,14] need to use the multi-axis robot and solve the forward and inverse kinematics of each joint angle value of multi-axis robot. They not only increase the complexity of the calibration, but also introduce more systematic error sources in the final calibration result and increases the measurement errors.

For the structured light measurement system, if there is no specific target, it will be difficult to calibrate the structured light plane. Thus,

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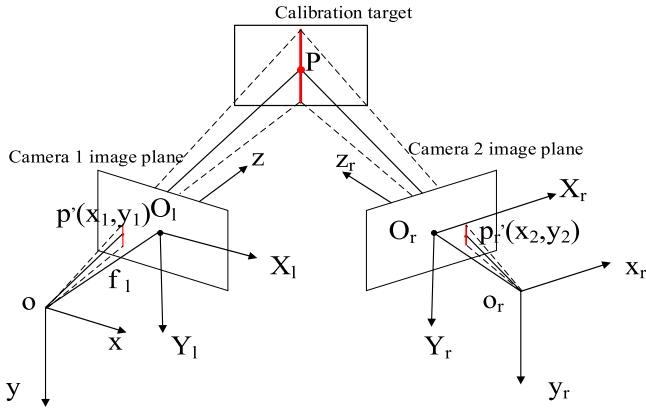


Fig. 1. The principle of the point clouds calculation.

the stereo vision calibration method (SVC) is proposed in this paper to calculate the parameters of the line structural light plane. The SVC method can use easily accessible objects (such as bottles, cans, and books) as targets rather than specific targets for calibration. It has two contributions: (1) The method uses the principle of stereo vision to calculate all point clouds projected on a target by structured light. Based on the principle of stereo vision, point clouds acquisition is not limited by the shape of the target. This method can use non-special targets to complete the calibration of the structured light plane. (2) The number of calibration images is reduced. This calibration method can be completed even only using single or double calibration images. After each calibration image is taken, all point clouds on the structured light can be calculated. Compared with the prior methods, more feature points are able to be obtained in a single shot with SVC. Therefore, the calibration can be completed with a smaller number of calibration images.

The remainder of this paper is organized as follows: Section 2 describes the algorithm principle of SVC method. The calibration experiments and accuracy verification using different targets are presented in Section 3. The final conclusion is drawn in Section 4.

2. Calibration algorithm principle

2.1. Computing the plane point clouds

The principle of the point clouds of the structured light plane on the target is shown in Fig. 1. P is a point on the line structured light, A and B are the position of P under the coordinate system of camera 1 and camera 2, respectively. The position of the image point of the P is (u_1, v_1) in the image plane of camera 1. The image point coordinate (u_2, v_2) of the P in the camera 2 image plane can be calculated by the principle of epipolar constraint.

According to the imaging model of the camera, Eqs. (1) and (2) are deduced as below:

$$\begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix} = \begin{bmatrix} f_1 & 0 & 0 & 0 \\ 0 & f_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} u_2 \\ v_2 \\ 1 \end{bmatrix} = \begin{bmatrix} f_2 & 0 & 0 & 0 \\ 0 & f_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \\ 1 \end{bmatrix} \quad (2)$$

where f_1 and f_2 are the focal lengths of camera 1 and camera 2, and the rotation matrix between camera 1 coordinate system $o - xyz$ and camera 2 coordinate system $o_r - x_r y_r z_r$ is

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}, \text{ and the translation matrix is } T = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}.$$

Then the relationship between (x, y, z) and (x_r, y_r, z_r) can be described as Eq. (3).

$$\begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \quad (3)$$

The point coordinate value in the point clouds of the line structured light projected on the target is described as Eq. (4):

$$\begin{cases} x = \frac{zX_1}{f_1} \\ y = \frac{zY_1}{f_1} \\ z = \frac{f_1(f_2t_x - X_2t_z)}{X_2(r_{31}X_1 + r_{32}Y_1 + f_1r_{33}) - f_2(r_{11}X_1 + r_{12}Y_1 + f_1r_{13})} \end{cases} \quad (4)$$

2.2. Point clouds processing and plane fitting

There are some noise points in the obtained plane point clouds, it is necessary to remove these noise points to fit a more accurate structured light plane. The process is as follows:

Firstly, the plane fitting of structured light is carried out, and the fitting algorithm is as the plane equation: $Ax + By + Cz + D = 0$, here set the a_0, a_1, a_2 as Eq. (5):

$$a_0 = -\frac{A}{C}, a_1 = -\frac{B}{C}, a_2 = -\frac{D}{C} \quad (5)$$

Then $z = a_0x + a_1y + a_2$.

Secondly, the degree of fitting between the fitted plane and the discrete points in point clouds is described as: $M = \sum_{i=1}^n (a_0x + a_1y + a_2 - z_i)^2$, the values of a_0, a_1, a_2 are calculated to minimize the M value and they are acquired by Eq. (6).

$$\begin{cases} \frac{\partial M}{\partial a_0} = 0 \\ \frac{\partial M}{\partial a_1} = 0 \\ \frac{\partial M}{\partial a_2} = 0 \end{cases} \quad (6)$$

Finally, the values of a_0, a_1, a_2 are calculated with Eq. (7), and then structured light plane is determined accordingly.

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^n x_i^2 & \sum_{i=1}^n x_i y_i & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i y_i & \sum_{i=1}^n y_i^2 & \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n y_i & n \end{pmatrix}^{-1} \begin{pmatrix} \sum_{i=1}^n x_i z_i \\ \sum_{i=1}^n y_i z_i \\ \sum_{i=1}^n z_i \end{pmatrix} \quad (7)$$

where n is the number of discrete points of the fitting plane used.

It is worth noting that when the fitting plane is completed, this fitted plane is not taken as the final result plane. Since there are many noise points in the plane point clouds, it is necessary to remove these noise points from the point clouds. The specific algorithm of noise reduction is as follows:

The distance of each point $P_1, P_2, \dots, P_N$ to the fitted plane is calculated by

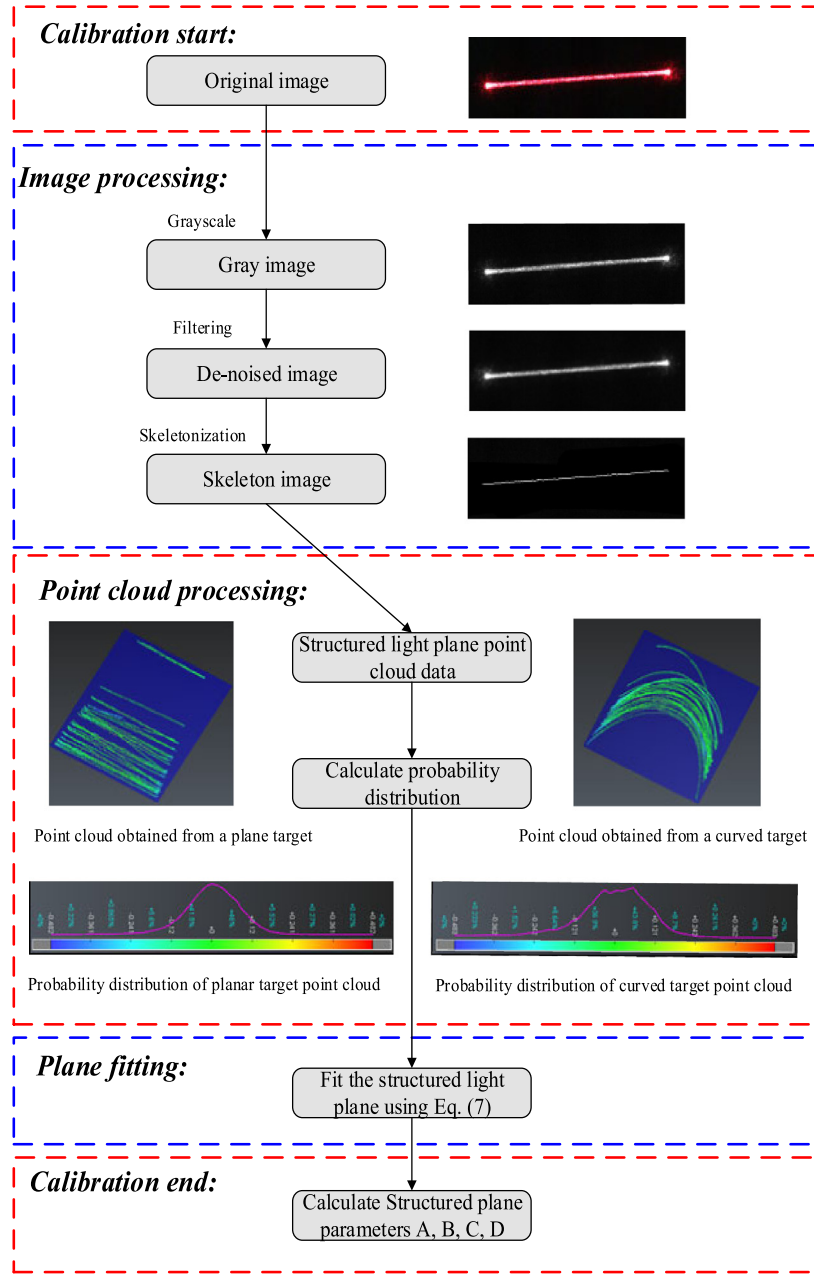


Fig. 2. The process of SVC method calibration.

$$d_k = \frac{|Ax_k + By_k + Cz_k + D|}{\sqrt{A^2 + B^2 + C^2}} (k = 1, 2 \dots n).$$

Assuming that the distance between the point on the positive direction of the plane normal and the fitted plane is positive, the distance between the point on the negative direction of the plane normal and the fitted plane is negative. Then the distance distribution curve is acquired, and its 95% confidence interval is calculated. The points outside the confidence interval are eliminated, and the remaining points are regarded as new fitting points which are used for fitting the result plane.

2.3. Image processing

Image processing is one of the key technologies to obtain the fitted point cloud, which includes image grayscale, image noise reduction and laser skeleton extraction (Fig. 2).

Image graying: The collecting the images gray out, at first. Since the wavelength of the laser is 650 nm, the red component weight in the

RGB channel will be readjusted to reduce the interference of ambient light. In the grayscale processing, the grayscale formula $Gray(i, j) = 0.8 * R(i, j) + 0.1 * G(i, j) + 0.1 * B(i, j)$ is used. Where $Gray(i, j)$ represents the gray value of pixel point (i, j) , and $R(i, j)$, $G(i, j)$, $B(i, j)$ respectively represent the red component, green component and blue component in RGB image.

Image denoising: In the process of imaging and digital transmission, the image inevitably generates various noises. The Gaussian filtering (adopting the 5×5 convolution) denoising algorithm is used to de-noise the image.

Extract laser skeleton: The width of the laser is generally 10–15 pixels. The laser intensity distribution is regarded as a Gaussian distribution, thus the laser skeletons are extracted by steger method [15] with a single pixel width.

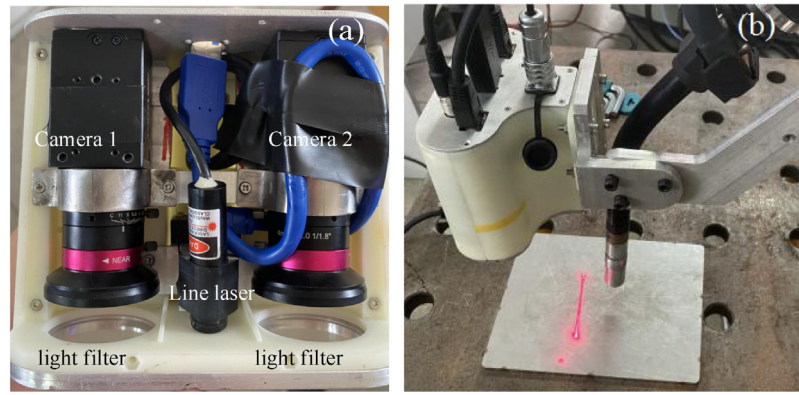


Fig. 3. Composition of structured light plane calibration system. (a) Calibration system composition. (b) The measuring system is installed on the robot.

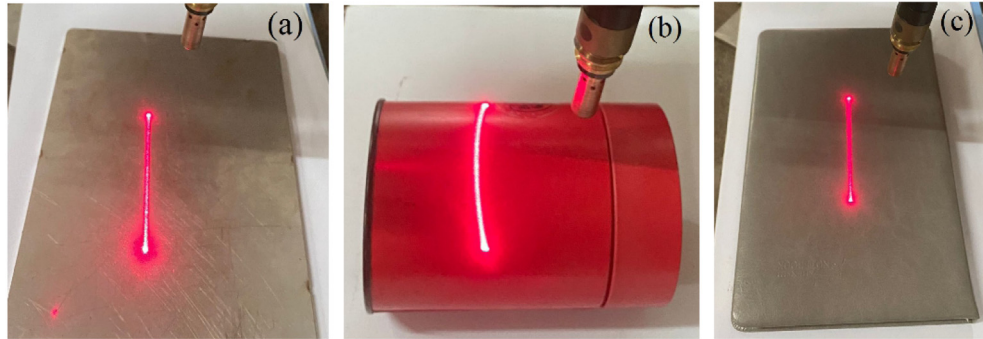


Fig. 4. Physical pictures of different targets as calibration objects. (a) 304 steel plate. (b) aluminum round can. (c) book.

2.4. Specific processing of calibration method

The target is placed in the camera's field of view in any posture, then the structured light project on the target and a calibration image is taken. The above process is repeated several times for collecting multiple calibration images.

Step 1: The camera is used to collect the image projected by the laser on the target, the laser skeleton is extracted by the image processing algorithm, and the point clouds data on the target are obtained based on Eq. (4).

Step 2: The target posture is changed, and then repeat Step1 several times (30 times in this paper) to obtain sets of point clouds of the structured light plane, and then mark the sets of point clouds as Cr .

Step 3: The method of least squares is used to fit the plane of the point clouds set, and mark the fitted plane as $P1$.

Step 4: Calculate the distances from each point $P_1, P_2, \dots, P_N$ in the point clouds Cr to the fitted plane $P1$, and then obtain the distribution function of point clouds with distance.

Step 5: The distance distribution can be regarded as a normal distribution, so the points that are away from the fitted plane $P1$ can be removed according to $\left| \frac{X-\mu}{\sigma} \right| > u_{0.25}$. The average value and variance of the distance between the point in the point clouds Cr and the fitted plane $P1$ are expressed by μ and σ respectively. The $u_{0.25}$ is the critical value corresponding to the 95% confidence level of the standard normal distribution.

Step 6: The points (within the 95% confidence interval) are used to fit the structured light plane with Eq. (7) and get the plane parameters.

The SVC method algorithm flow is shown in Fig. 2.

3. Experiment

3.1. Calibration system description

The calibration system consists of a computer, two CCD cameras, several filters, and a line laser emitter (Fig. 3). The two CCD cameras

and the laser emitter are fixed by the support structure. The filter is installed in front of the camera lens to reduce the interference of ambient light. The computer CPU model is Intel(R) core(TM) i7-8750H. CPU operating frequency is 2.20GHZ. The model of the two cameras with a resolution of 2592×1944 pixels is DFK33GP006. And the laser emitter model is FUZHE 650AB100-GC16 (Fig. 3a).

The internal reference matrix K_1 of camera 1 is:

$$\begin{bmatrix} 1977.3 & 0 & 1386.3 \\ 0 & 1981.5 & 962.3086 \\ 0 & 0 & 1 \end{bmatrix}$$

The internal reference matrix K_2 of camera 2 is:

$$\begin{bmatrix} 1891.4 & 0 & 1274.9 \\ 0 & 1894.9 & 950.9911 \\ 0 & 0 & 1 \end{bmatrix}$$

The external parameter matrix of camera 1 and camera 2 is:

$$\begin{bmatrix} 0.9980 & -0.0097 & 0.0630 & -113.9751 \\ 0.0102 & 0.9999 & -0.0082 & -3.1351 \\ -0.0629 & 0.0088 & 0.9980 & 13.4226 \end{bmatrix}$$

The line laser emitter projects the laser on the target (Fig. 3b). And then, images are captured by computer-controlled cameras. After that, the images will be transmitted back to the computer through the network cable for image processing.

3.2. Calibration experiments

Three items are used as targets to calibrate the structured light plane (Fig. 4). The flat targets with smooth and rough surfaces are

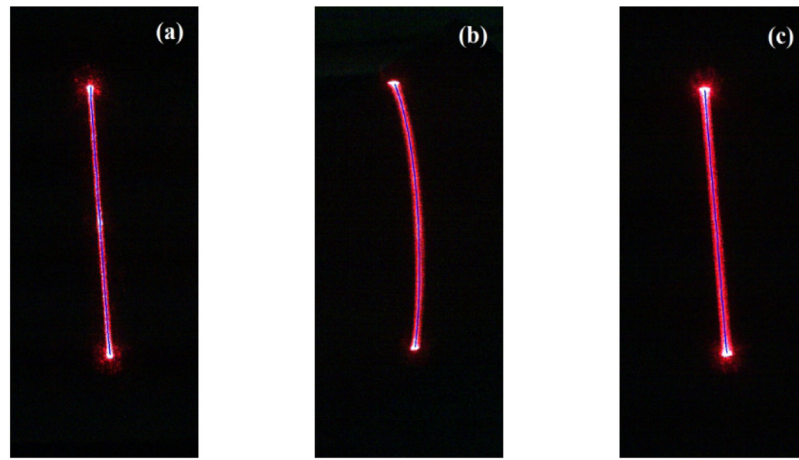


Fig. 5. Laser skeletons were extracted from different targets. (a) 304 steel plate target. (b) Round can target. (c) book target.

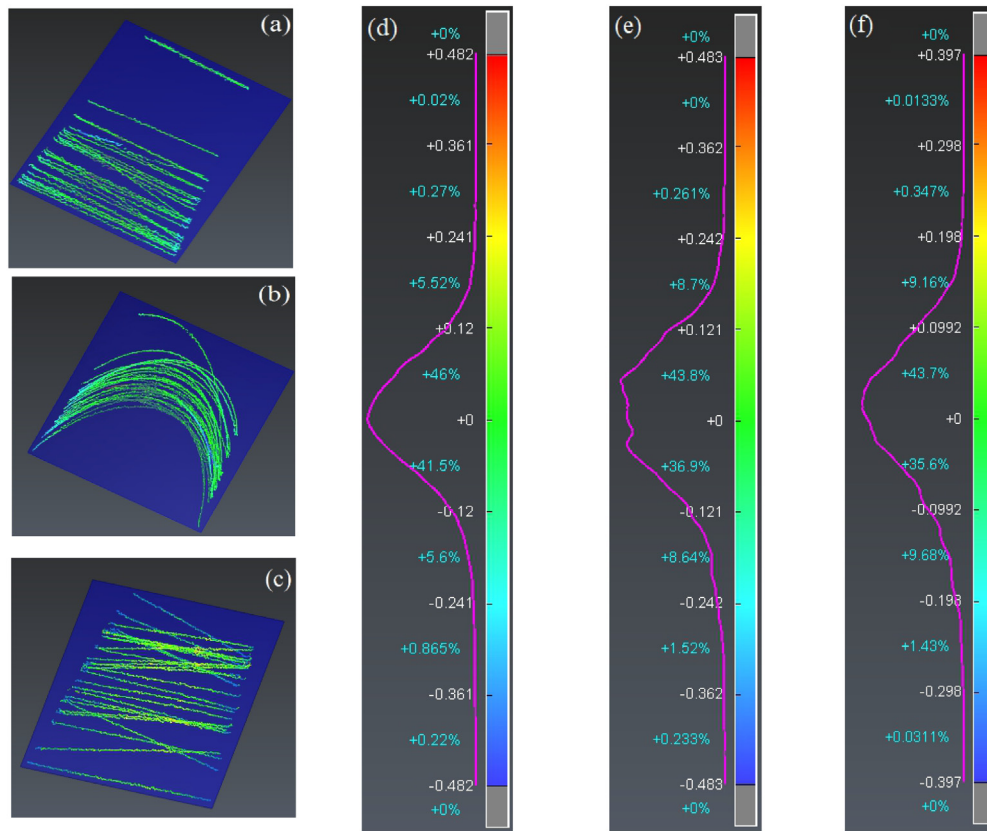


Fig. 6. Plane point clouds diagram and probability distribution of three targets in 30 different postures. (a) The fitting planes calibrated with 304 steel plates. (b) The fitting planes calibrated with aluminum cans. (c) The fitting planes calibrated with books. (d) The distribution diagram of the distance from the point clouds to the fitted plane with 304 steel plate as the calibration object. (e) The distribution diagram of the distance from the point clouds to the fitted plane with aluminum can as the calibration object. (f) The distribution diagram of the distance from the point clouds to the fitted plane with book as the calibration object.

represented by 304 steel plate and book, respectively. The aluminum round is regarded as a curved target with a smooth surface.

The laser is projected on the targets. The Steger method is used to extract the laser skeleton images of different targets. Laser skeletons were extracted and drawn with blue lines (Fig. 5).

It can be seen that the laser fringe of 304 steel plate is thinner due to the mirror reflection of laser projection on 304 steel plate. On the contrary, it has lower brightness and coarser fringe on book, because of the diffuse reflection produced by laser projection on the surface. The laser streak projected on the round can appears curved in shape compared to the plane targets. This indicates that the Steger method

can extract the excitation skeleton correctly no matter what target is used. Here, the R^2 , sum of squares due to error (SSE) and mean of residuals (MR) of the fitted line [16,17] were used to illustrate the effect of Steger method in extracting sub-pixel laser skeleton. If the R^2 is closer to 1, it means that the higher the stability of strip extraction. If the value of SSE is closer to 0, it shows that the curve fitting degree is better. In addition, the smaller the mean of residual error, the higher the accuracy of skeleton extraction. Ten laser skeleton images of 304 steel plate were selected to calculate the averages of R , SSE, and MR (Table 1). The value of R^2 is 0.9971 and the average residual is about 0.16 pixels.

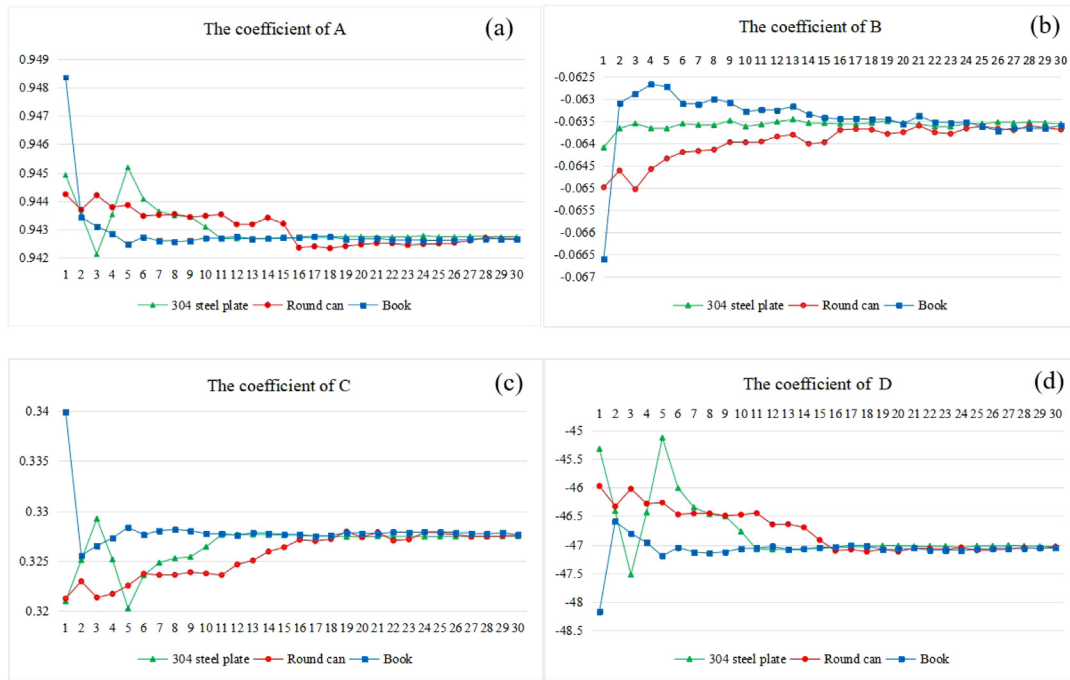


Fig. 7. Four coefficients diagram of structured light plane equation. (a) Diagram of coefficient A under multiple experiments. (b) Diagram of coefficient B under multiple experiments. (c) Diagram of coefficient C under multiple experiments. (d) Diagram of coefficient D under multiple experiments.

Table 1
Steger method of laser skeleton fitting results.

The fitting equation	R^2	SSE	MR
$y = 11.822x - 13390.258$	0.9971	52.9401	0.16 pixels

Fig. 6(a)–(c) show the point clouds data of three targets in 30 different postures. The distribution of the distance between each point in the point clouds and the fitting plane is shown in Fig. 6(d)–(f). According to the distribution, the distance interval of 95% confidence interval is calculated. Then the points in the confidence interval are used for fitting the structured light plane. The confidence intervals and calibration results of the three targets are shown in Table 2.

Fig. 7 shows the changing trend of coefficients A, B, C, and D in the structured light plane equation with the increase of the calibrated image number. These plane coefficients (A, B, C, and D) gradually converge because they calibrate the same structured light plane. Using a book as a target for calibration, the plane coefficients tend to be stable when the number of images reaches four. However, when the images number of testing 304 steel plate and aluminum reach about 14 and 19 respectively, the coefficients can be stable.

4. Results and discussion

Three workpieces of different sizes with manufacturing accuracy of 0.01 mm and a checkerboard with manufacturing accuracy of 0.005 mm were used for accuracy verification. The dimensions of the three workpieces are 30 mm, 40 mm, and 50 mm respectively, and the single size of the checkerboard grid is 20 mm. In order to calculate measurement errors, the values obtained by measuring the size of workpieces and checkerboard are compared with the nominal values.

4.1. Measurement precision analysis of checkerboard size

For checkerboards, project the laser on the edge of the checkerboard grid to calculate $D_1 = |Q_1Q_2|$, $D_2 = |Q_1Q_3|$, $D_3 = |Q_1Q_4|$ (Fig. 8b). The nominal values should be 20 mm, 40 mm and 60 mm.

When the book is used as the target and only 4 calibration images are used, the average measurement error has converged to about 0.079 mm (Fig. 9). By comparison, when the 304 steel plate is used as the target and the number of calibration images is about 14, the average measurement error is about 0.071 mm. The average measurement error converges to about 0.069 mm when the round can is used as the target and the number of calibration images reaches 19. In addition, no matter what kind of target is used for calibration, the average measurement errors gradually decrease as the number of calibration images increases, and the final average measurement error is about 0.065 mm (Fig. 9d).

4.2. Measurement accuracy analysis of square workpieces size

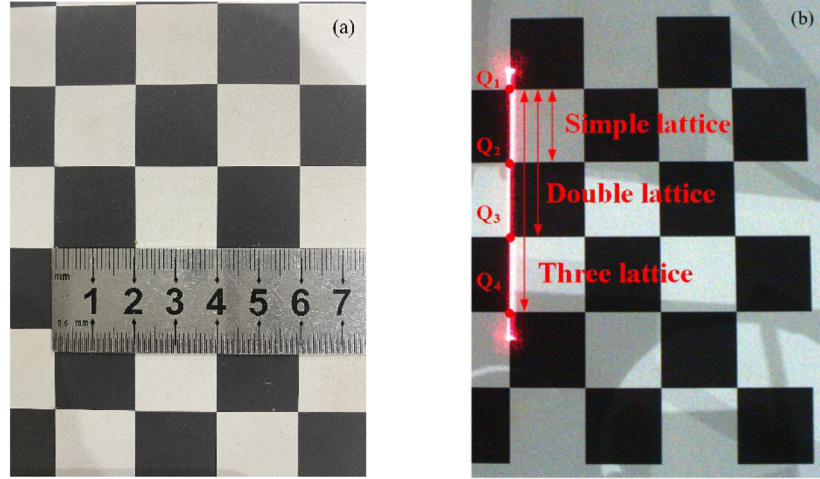
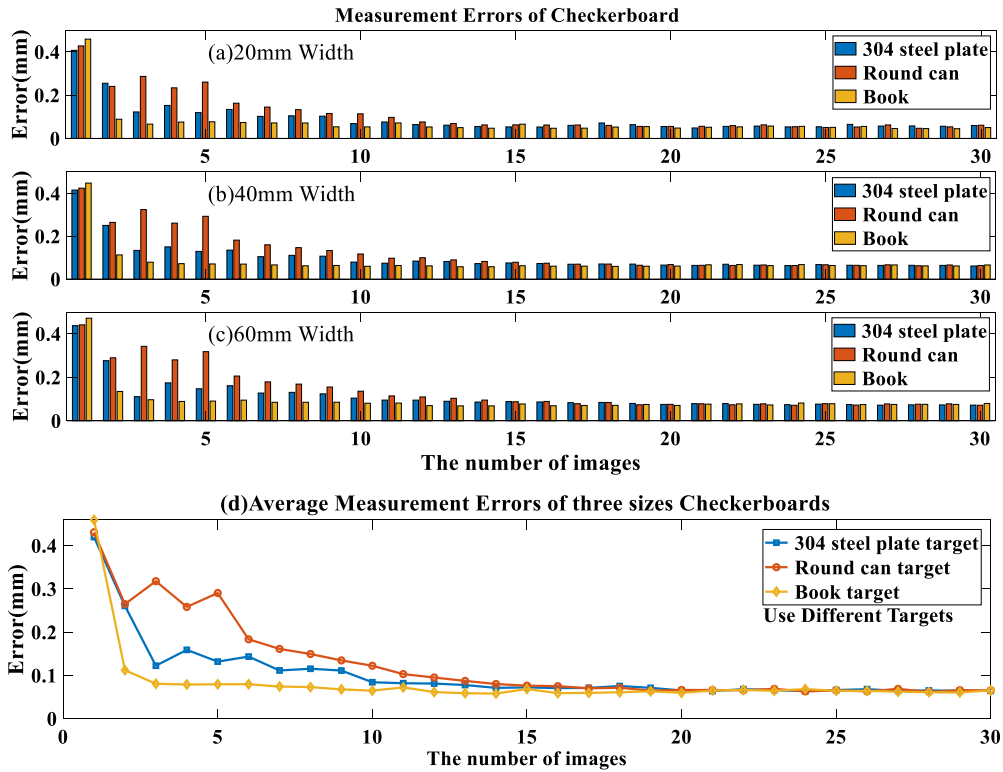
In addition to measuring the size of the checkerboard, we also measured three square workpieces of different sizes. The nominal side lengths of the workpieces are 30 mm, 40 mm, and 50 mm. The laser is projected on the edge of the workpiece, two characteristic points Q_1 and Q_2 (Fig. 10) are formed, the distance $D = |Q_1Q_2|$ is calculated, and the values of measurement errors are calculated from the nominal values of the workpieces.

The results of measuring workpieces size are similar to those of measuring the checkerboard (Fig. 11). The error convergence rate is higher using book target, and the average error tends to be stable after using only four calibration images. The next is 304 steel plate, the average error tends to be stable after using 14 calibration images, and the average error is about 0.080 mm. For aluminum round target, the convergence speed of average error is the slowest. And it only converges when the calibration image number reaches 19, and the average measurement error is about 0.076 mm. However, the average measurement error of different objects is very close after using 30 calibration images, at about 0.071 mm (Fig. 11d). Traditional laser plane calibration methods are usually based on cross ratio invariability and using specific targets for calibration [9–12]. There is a results comparison between the SVC method and four traditional calibration methods in Table 3. The calibration pixel error of the SVC method is about 0.24 pixels ($\frac{k_x \times e}{d} = 0.24$, where $k_x = 1977.3$ represents the value of the first row, first column of the camera intrinsic matrix, $e = 0.068$

Table 2

The plane equations calibrated with 304 steel plates, aluminum cans and books respectively obtained through a series of processing.

Target type	Number of point clouds	Number of fitting points	Confidence interval(mm)	The fitted plane equation
304 Plate	19 994	18 994	(−0.140,0.137)	$0.942722x - 0.063563y + 0.327485z - 47.0309 = 0$
Round Can	20 964	19 916	(−0.227,0.193)	$0.942626x - 0.063689y + 0.327546z - 47.0358 = 0$
Book	21 478	20 404	(−0.154,0.129)	$0.942649x - 0.063601y + 0.327624z - 47.0586 = 0$

**Fig. 8.** Checkerboard measurement errors experiment (a) Checkerboard physical picture. (b) Checkerboard size measurement image.**Fig. 9.** Measurement errors based on the size of different checkerboard cells (the side length of 1 grid is 20 mm). (a) Measurement errors of a square workpiece with a side length of 20 mm. (b) Measurement errors of a square workpiece with a side length of 40 mm. (c) Measurement errors of a square workpiece with a side length of 60 mm. (d) Average measurement errors.

represents the average measurement error, and $d = 500$ represents the height of the laser projection on the target).

In addition, compared with previous calibration methods, the proposed SVC method can significantly reduce the calibration complexity. Through the analysis of the plane calibrated by using different targets,

we found that SVC method can obtain high precision calibration results with non-specialized targets easily, and the measurement error of rough plane target converges fastest. The minimum average measurement error is about 0.068 mm, which proves that this method can use non-specialized targets to complete structured light plane calibration. But

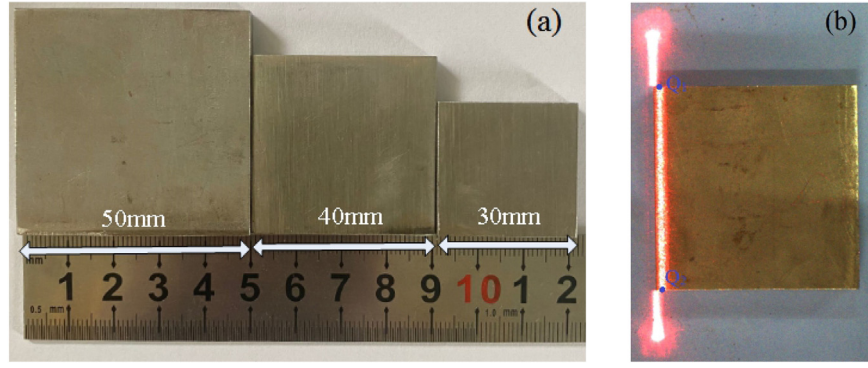


Fig. 10. Workpieces size measurement errors experiment (a) Workpieces dimension measurement image. (b) Laser calibration image.



Fig. 11. Measurement errors of square workpieces of three different sizes. (a) Measurement errors of a square workpieces with a side length of 50 mm. (b) Measurement errors of a square workpieces with a side length of 40 mm. (c) Measurement errors of a square workpieces with a side length of 30 mm. (d) Average measurement errors.

Table 3

Comparison with other calibration methods.

No.	Calibration target	Calibration pixel error
1	Ball target [9]	0.22 pixels
2	parallel straight lines target [10]	0.35 pixels
3	Checkerboard target [11]	0.87 pixels
4	Checkerboard target [12]	0.57 pixels
5	Non-specific target	0.24 pixels

the convergence speed of the measurement errors is not the same by using different targets. Using a flat target with a rough surface, the measurement errors will converge quickly. Its measurement error is about 0.080 mm only by 4 calibration images, and when 30 calibration images are used, the measurement errors reduce to 0.068 mm. For the smooth plane target, the average measurement error is about 0.076 mm and convergence while using 14 or more than 14 calibration images. However, the error convergence speed is the slowest with curved targets. the measurement error is about 0.071 mm with more than 19

calibration images. These results indicated that the users can complete the structured light plane calibration with easily available objects. The calibration can even be completed with only one calibration image to meet the precision requirements (0.4 mm–0.5 mm), as the precision requirement of the measurement system is not very high (such as railway length measurement). In addition, it is recommended to use a flat object with the rough surface instead of a special target for structured light plane calibration.

5. Conclusion

In this paper, a method of structured light plane calibration using non-specific targets is proposed. Firstly, the image of line structured light fringe projected on the target is collected, and the center line of laser stripe is extracted by image processing. Secondly, multiple groups of point clouds on the plane are obtained by changing the target posture continuously based on the principle of stereo vision, and the points with large errors are removed by the point clouds

filtering algorithm. Finally, the calibration is completed by fitting the structured light plane. Three kinds of non-specific targets (304 steel plate, aluminum round can, and book) were used to complete the calibration experiment. The influence of the number of calibration pictures, target material, and target shape on the calibration results was explored. It is proved by measuring three different sizes of workpieces that when the number of calibration images reached about 27, the accuracy would not be improved with the increase of image number. Compared with the classical plane calibration method, SVC method can not only calibrate with easily available objects but also improve the calibration accuracy. In the future, higher calibration accuracy can be achieved by improving the point clouds processing algorithm or the plane fitting algorithm. In addition, more different targets can be used to explore the impact of target shapes and materials on the calibration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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